Daliparthy Ravi Krishna Charan

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Summary

Software engineer and published researcher with hands-on experience building large-scale deep learning systems, full-stack applications, and analytical dashboards. Peer-reviewed publication at IIT ISM Dhanbad; national AI/ML hackathon finalist (IIT Tirupati & Ministry of Panchayati Raj). Strong problem-solving foundation with active competitive programming practice.

Education

Sastra Deemed University — Oct 2022 – Jun 2026
*B.Tech. Information & Communication Technology — CGPA: **8.5444** / 10.0 — Top 10% of the Department* — *Thanjavur, India*

Narayana Junior College — Apr 2020 – Jun 2022
*Intermediate – Mathematics, Physics, Chemistry — **97.8%*** — *Hyderabad, India*

Technical Skills

Languages: Python, C, C++, Java, SQL, JavaScript (TypeScript), HTML, CSS

ML & Data: TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Transformers (HuggingFace), Power BI, DAX

Full-Stack: React.js, Angular 16, Node.js, Express.js, MongoDB, MySQL, REST APIs, Git

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, OS Concepts, System Design

Projects

Rooftop Material Segmentation from Drone Imagery — ***GitHub*** — *Python | TensorFlow | HuggingFace | OpenCV*

- Benchmarked 8 segmentation architectures on **17,954 geo-referenced buildings** in Nacala, Mozambique for malaria-risk mapping; countered a **56:1 class imbalance** with Focal and Tversky losses and a novel MegaEnsemble with Deep Ordinal Watershed decoding, reaching **IoU 0.867** and **AP50 0.840**.

Expense Tracker — ***GitHub*** — ***Live*** — *Angular 16 | Node.js | Express.js | SQLite | Turso*

- Engineered an offline-first expense platform with WAL-mode SQLite for in-process reads and Turso for cloud persistence; used idempotent batch sync to prevent duplicate writes and **integer-cent arithmetic** to eliminate floating-point errors. Secured with JWT and Zod, validated by **34 automated tests**, and shipped as an installable PWA.

Resume Agent – AI Resume Optimizer — ***GitHub*** — ***Live*** — *React 19 | TypeScript | Cloudflare Workers | Workers AI*

- Architected an edge-native agent on **Llama 3.3 70B** that runs resume tailoring and ATS fit scoring in parallel, with a validation-and-retry quality gate and zero-fabrication guardrails; Durable Objects with SQLite preserve per-session state over WebSockets, and the pipeline exports PDFs with optional Gmail delivery.

AI Technical Interviewer — ***GitHub*** — ***Live*** — *Next.js | FastAPI | Python | TypeScript | Groq API*

- Built an adaptive interview platform on Groq LLMs with automatic model fallback that cross-examines candidates on their own PDF resumes and benchmarks questions against **25+ company interview bars**; includes a **7-language** code sandbox with Big-O analysis and an evidence-backed scorecard.

Achievements & Research

Peer-Reviewed Publication – *Intelligent Seismic Activity Detection: A Deep Learning Approach*; Centenary National Seminar on Strong Motion Earthquake (SME-2025), **IIT (ISM) Dhanbad** — Mar 2025

National AI/ML Hackathon 2026 – Finalists — *GitHub* – Organised by IIT Tirupati & Ministry of Panchayati Raj; developed automated DTM generation and drainage network design from drone point-cloud datasets — Jun 2026

Certifications

Matrimorphosis – Campus to Corporate Program – Matrimony.com; **selected among 250 from 3,000+ applicants**; 5-month industry readiness program — Dec 2024–May 2025

TATA iQ GenAI Powered Data Analytics Simulation – Forage; AI-powered data analytics and strategy development for Financial Services — Jun 2025

Advanced Data Science & Machine Learning – NilTech-Edu — Jul–Dec 2024

Virtual Internship – Sync Interns — Nov–Dec 2023

Leadership

President, Maths Club – Sastra Deemed University; led planning and execution of academic events, workshops, and cross-functional team coordination — Dec 2022— Jun 2026

Core Member, Robotics Club – Sastra Deemed University; promoted hands-on robotics learning through technical sessions — Apr 2023–May 2024

Core Member, Daksh Workshops – Assisted in organising career guidance workshops and managing large-scale events — Mar–Nov 2024